Database Lab : Read Replica Stalled Replication Escalation Table

Duration : 30 Min.

Scenario

As a Database Platform Engineer at NIIT, a disaster recovery replica has stalled. You must create an incident escalation table in MySQL and record time-series lag markers and root causes to notify the platform on-call team.

Input Details

The environment has been pre-configured with the following local database platform instances:

- **Database Host:** `localhost` (Ubuntu Linux VM)
- **MySQL Service:** `localhost:3306` | DB: `mysql` | User: `root` | Password: `MySQLPassword123!` | CLI: `mysql`

Username & Naming Conventions

Your candidate prefix is: **`<prefix>`** (e.g. `student-exam123`).

Resource naming: - **MySQL Table:** `<prefix>_replica_escalation` (in database `mysql`)

Task Objectives

Perform the following configuration steps directly via `mysql` CLI:

1. Create Replication Escalation Table

- In MySQL, create table `mysql.<prefix>_replica_escalation` with columns:
- `ticket_id VARCHAR(30) PRIMARY KEY`
- `replica_id VARCHAR(50) NOT NULL`
- `lag_seconds INT NOT NULL`
- `stall_status VARCHAR(30) NOT NULL`
- `priority VARCHAR(10) NOT NULL`
- `escalated_by VARCHAR(50) NOT NULL`

2. Record Incident Escalation Data

- Insert an escalation entry: `INSERT INTO mysql.<prefix>_replica_escalation VALUES ('INC-84910', 'replica-dr-01', 2700, 'STALLED_REPLAY', 'P1', '<prefix>');`

3. Verify Escalation Query

- Run `SELECT * FROM mysql.<prefix>_replica_escalation WHERE priority = 'P1';`.

Verification

Once completed, run the verification script:

```
python run.py
```

The driver connects to MySQL on `localhost:3306` and validates the table.

Grading Criteria

Your performance will be evaluated based on the following test cases:

Test Case	Requirement	Marks
TC1	MySQL Table (_replica_escalation) Creation	3 Marks
TC2	Replication Stall Metrics (Lag Seconds, Status) Population	3 Marks
TC3	Severity Priority (P1) Categorization	2 Marks
TC4	Escalation Metadata & Candidate Prefix Verification	2 Marks

Total Score: 10 Marks

Important Notes

- Replace `<prefix>` with your candidate prefix.
- All commands must be run in the live mysql console.